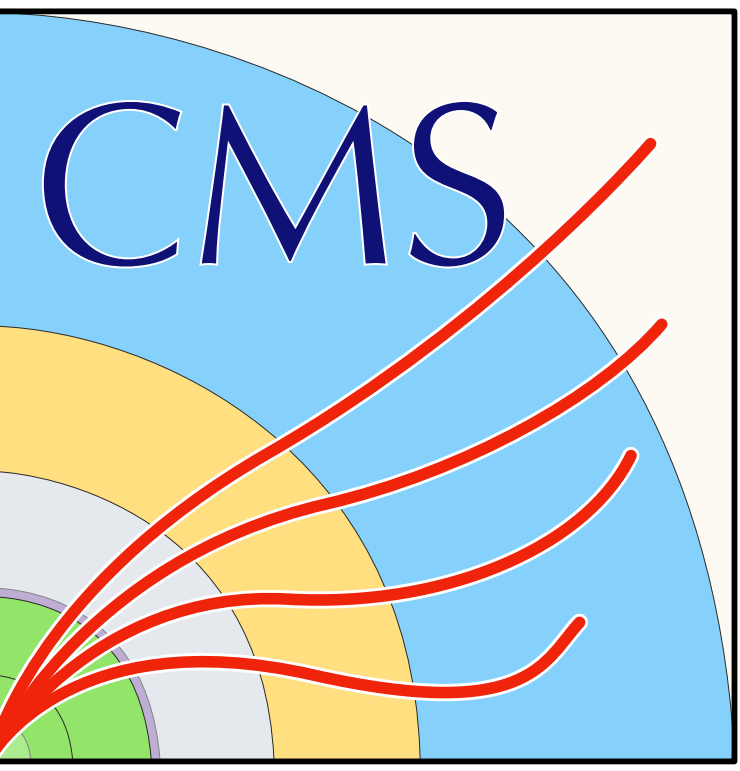




4-Lepton Off-shell Higgs boson Production Measurements

Mohit Srivastav, HIG-25-003 Authors

Johns Hopkins University, CMS Joint 4-Lepton Study Team (CJLST)



The off-shell Higgs boson

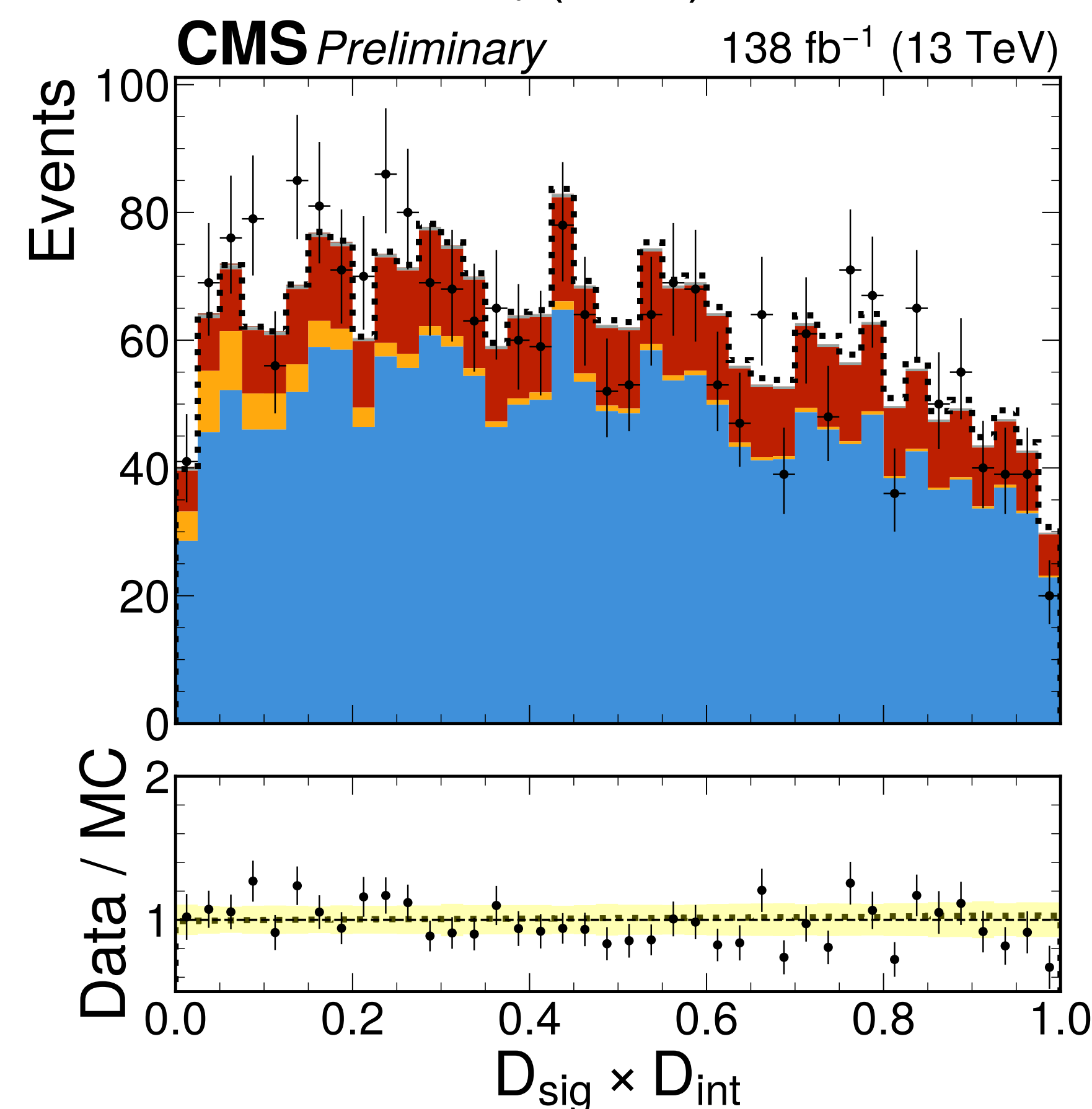
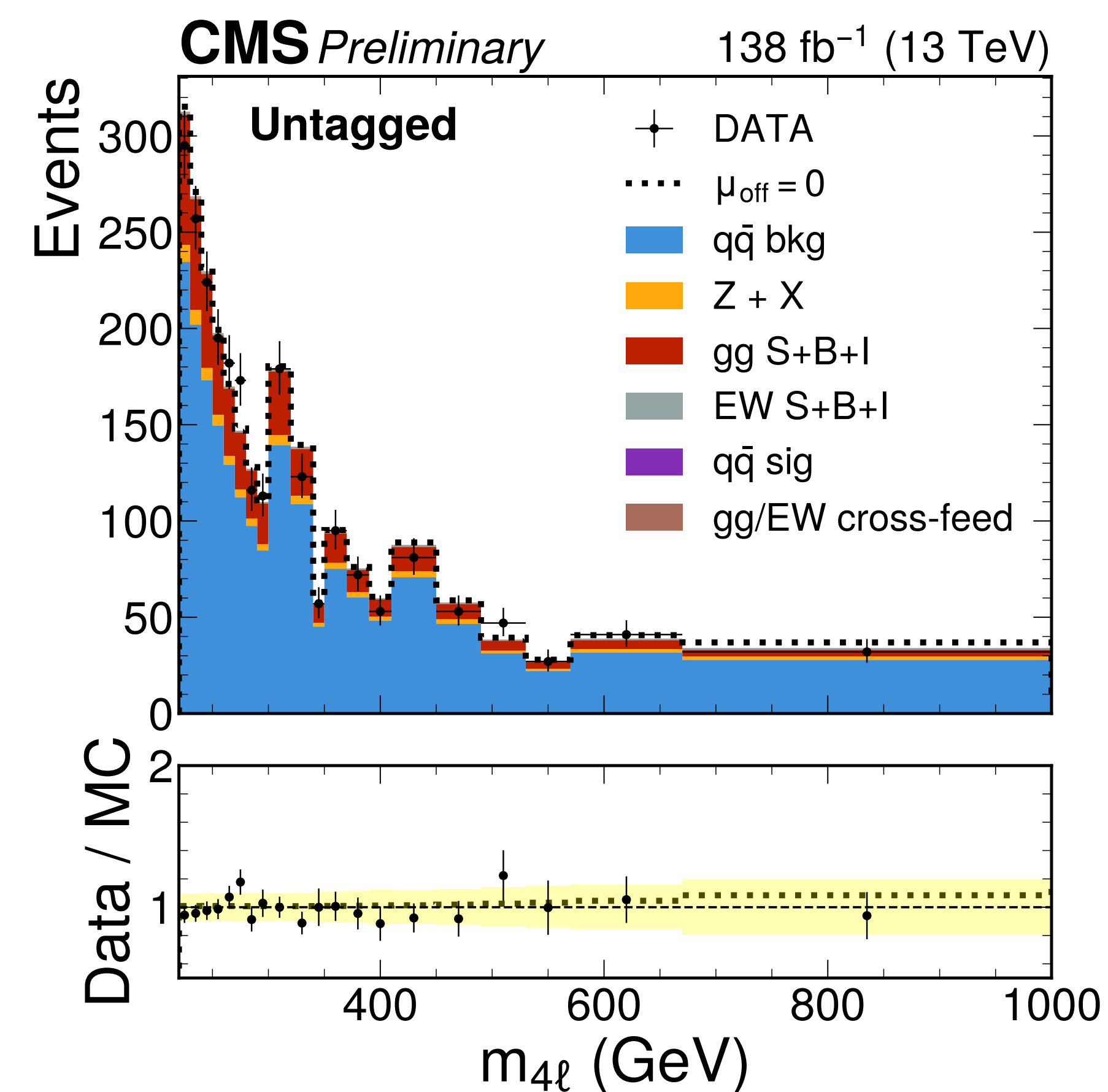
The off-shell region of the Higgs boson is enhanced in VV decays above $2m_V$, and is characterized by a strong interference component. The hallmark of the off-shell region is the independence from the width,

$$\frac{\sigma(i \rightarrow H \rightarrow f)}{ds} \propto \frac{1}{(s - M_H^2)^2 + M_H^2 \Gamma_H^2} \quad (1)$$

$(\sqrt{s} - M_H) \gg \Gamma_H$ when $\sqrt{s} \geq 2m_V$

Observables

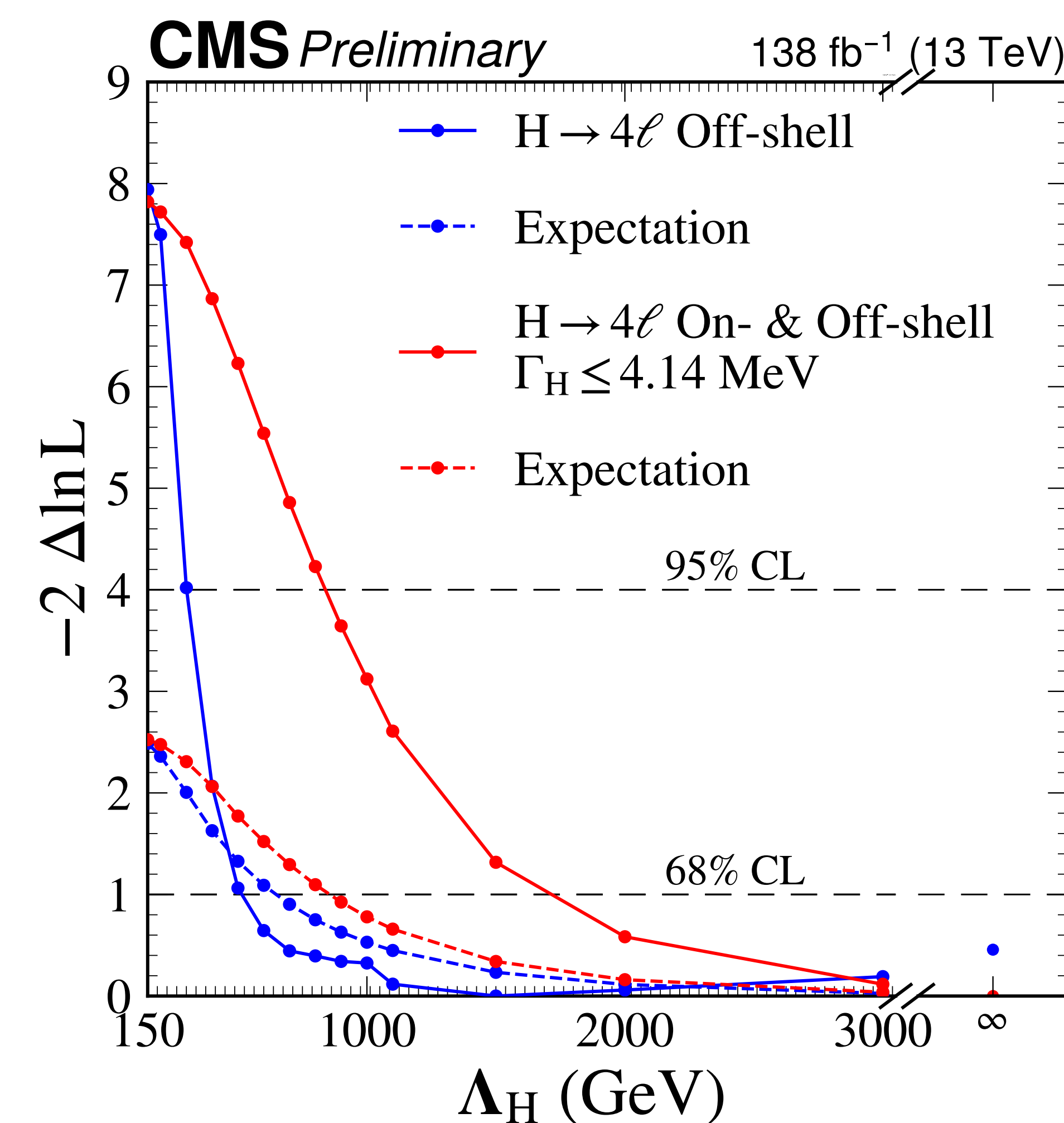
Matrix-element discriminants, together with $m_{4\ell}$, were used as observables.



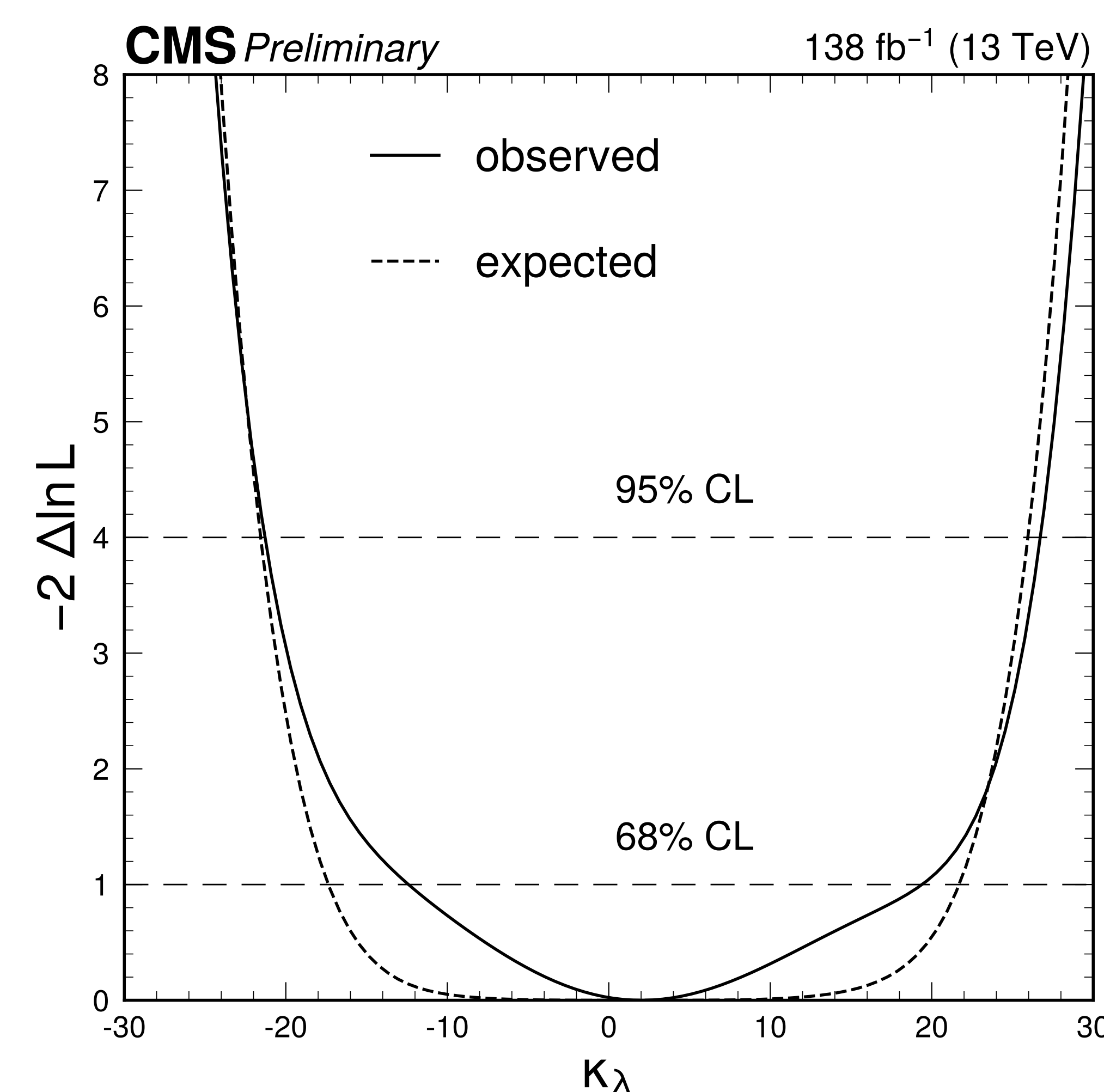
Higgs Substructure

We probe a nucleon form factor through the threshold energy Λ_H that tests Higgs boson substructure.

$$F(q^2) = \left[\frac{1}{1 + |q^2|/\Lambda_H^2} \right]^2, \quad d = \frac{\hbar c}{\Lambda_H}$$

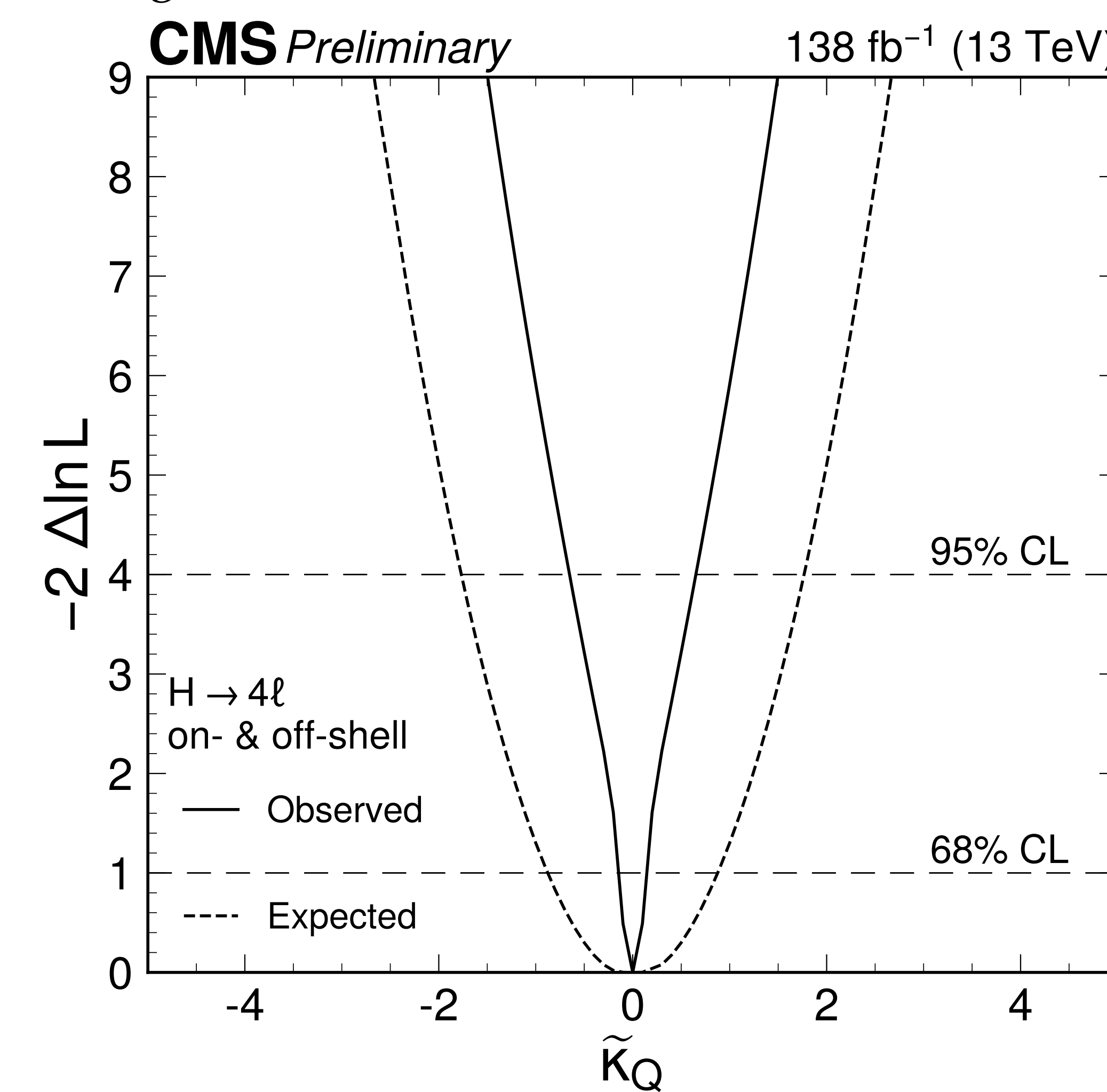


Higgs Self Coupling

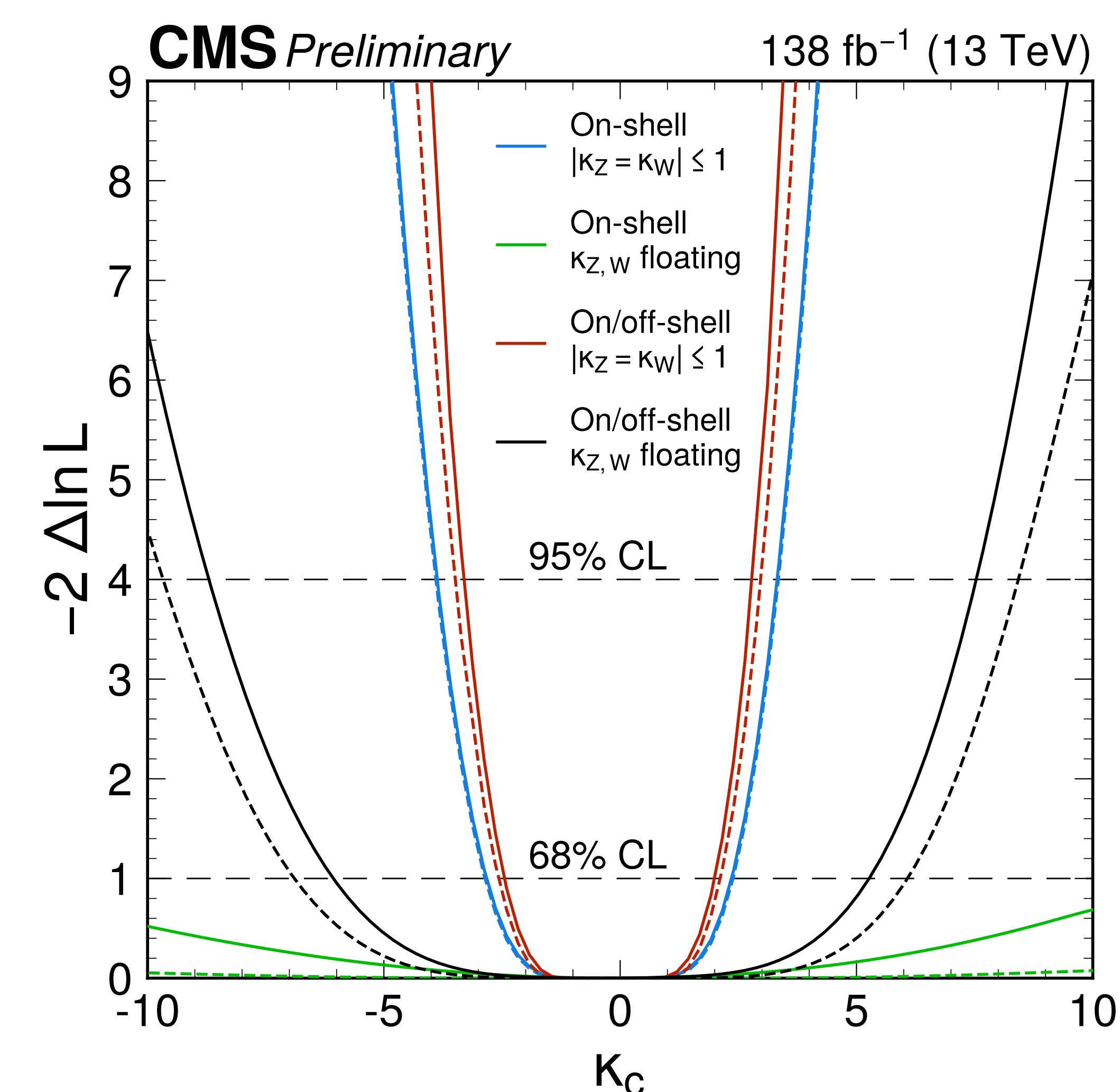


Yukawa Couplings

Scans over $\kappa_{tQ}, \tilde{\kappa}_{tQ}$ in the EFT limit for gluon-fusion production, where $\kappa_Q \propto C_{HG}$ and $\tilde{\kappa}_Q \propto \tilde{C}_{HG}$. CP-odd sensitivity comes from interference (or lack thereof) with SM background in off-shell.

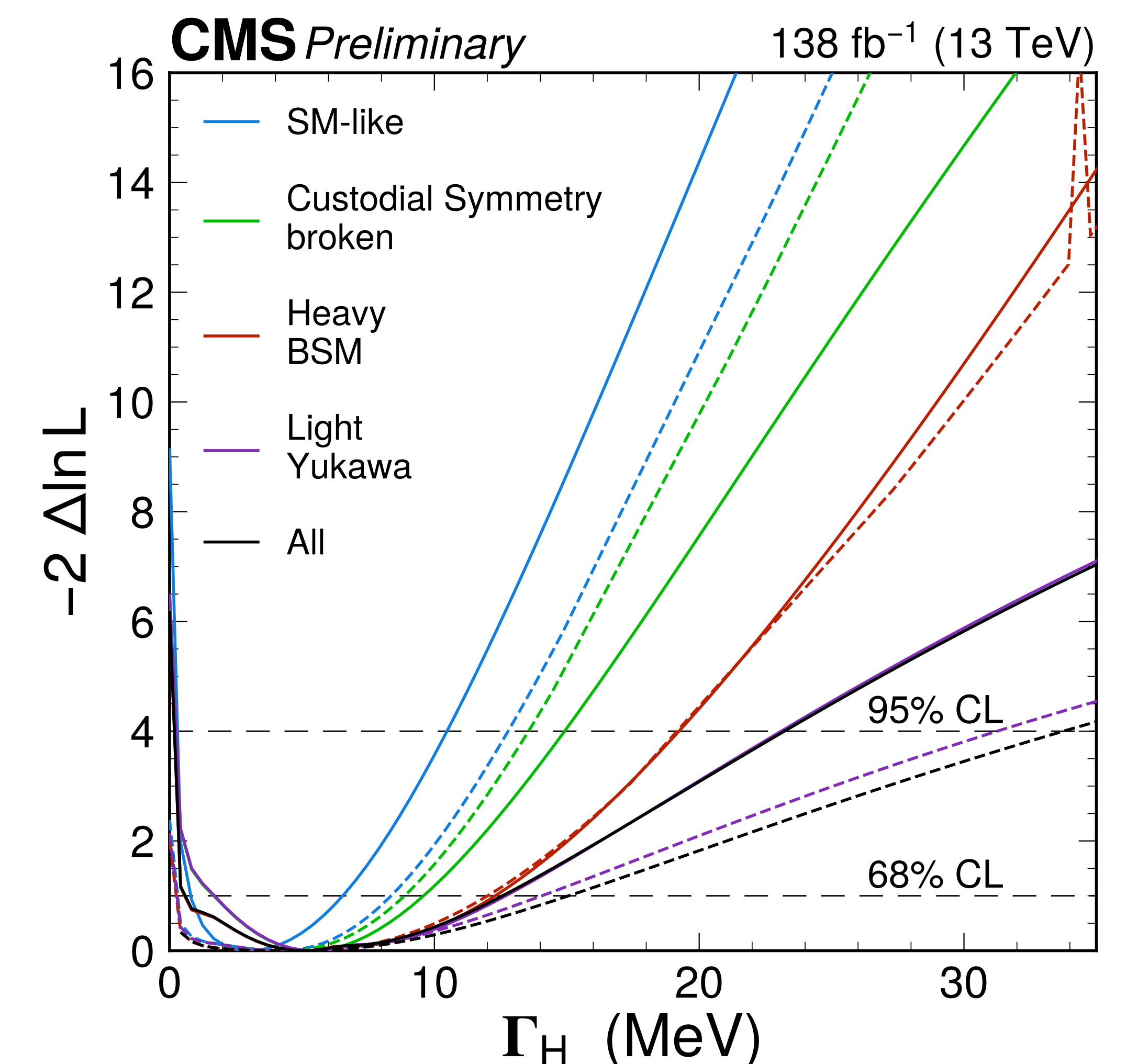


Scans over u,d,s, and c quarks build off of previous work and utilize off-shell to remove constraints on vector boson couplings.



Anomalous Higgs width

Anomalous terms are allowed to float simultaneously to measure their effects on Γ_H .



Run-2 Off-shell SM Combination

A combination of vector-boson decay channels is performed for all the Run-2 off-shell analyses conducted to retrieve the most precise measurement of the SM width of the Higgs boson to date.

